

Sinclair ZX Spectrum

SINCLAIR BASIC · TIMEX SINCLAIR 2068 · Z80A · 256×192 · 8 COLOURS · BEEP

Keyword-per-key entry like the ZX81: cursor shows **K** keyword, **L** letter, **C** caps, **E** extended (colours/tokens), **G** graphics. Colour is per 8×8 cell. The **Timex 2068** is a Spectrum superset with AY sound, joysticks and a cartridge dock.

PROGRAM CONTROL

RUN	Start; RUN 100 from a line
LIST	LIST 10 from a line
NEW	Erase program
CLEAR [n]	Clear vars / set RAMTOP
CONTINUE	Resume after STOP / BREAK
STOP	Halt program
SAVE "n"	Save (see Tape & Files)
LOAD "n" VERIFY	Load / check from tape
MERGE "n"	Load program, keep current

BASIC STATEMENTS

LET	Assign a value
PRINT ?	; joins, , tabs
PRINT AT r,c	Row 0-21, col 0-31
INPUT	Prompt & read (LINE for strings)
IF...THEN	Branch (statement after THEN)
FOR...NEXT	FOR I=1 TO 9 STEP 2
GOTO GOSUB	Jump / call; RETURN
READ DATA	RESTORE rewinds
DIM	DIM A(20), A\$(5,8)
REM RANDOMIZE	comment / seed RND
DEF FN	Define a function
POKE a,v OUT p,v	Write byte / I/O port
PAUSE n	Wait n frames (50 = 1 s)

FUNCTIONS

ABS SGN INT	magnitude/sign/floor
SQR SIN COS TAN	root, trig (radians)
ASN ACS ATN LN EXP	inverse trig / log / e^x
PI RND	3.14159 / random 0-1
PEEK(a) IN(p)	read byte / read port
LEN CODE VAL VAL\$	string ops
STR\$ CHR\$	number↔char
INKEY\$	Scan keyboard (no wait)
ATTR(r,c)	Colour attribute of a cell
POINT(x,y) SCREEN\$(r,c)	pixel set? / char at cell
BIN FN USR	binary / call FN / call ML

COLOUR & ATTRIBUTES

0-7

INK n	Foreground colour 0-7
PAPER n	Background colour 0-7
BORDER n	Screen border colour
BRIGHT 0/1	Normal / bright
FLASH 0/1	Swap ink↔paper periodically
INVERSE 0/1 OVER 0/1	Swap / merge when printing

Colour is per 8×8 cell (the famous 'attribute clash'). ATTR(r,c) reads it back.

COLOUR SET

INK / PAPER

0 Black	1 Blue
2 Red	3 Magenta
4 Green	5 Cyan
6 Yellow	7 White

Add 8 to a colour value (or BRIGHT 1) for the brighter shade.

GRAPHICS & SOUND

256×192

PLOT x,y	x 0-255, y 0-175 (origin bottom-left)
DRAW dx,dy[,a]	Line, relative; a = arc angle
CIRCLE x,y,r	Circle
POINT(x,y)	1 if pixel set
CLS	Clear screen
BEEP d,p	Sound: duration s, pitch (semitones from C)

TAPE & FILES

SAVE "n"	Save BASIC program
SAVE "n" LINE k	Auto-run from line k on load
SAVE "n" CODE a,l	Save memory block
SAVE "n" DATA a()	Save an array
LOAD "n" CODE / SCREEN\$	Load block / screen image
VERIFY MERGE	Check / merge program
LOAD "" loads the next program. Microdrive/disc add CAT FORMAT ERASE MOVE OPEN# CLOSE#.	

CURSOR / ENTRY MODES

keyword-per-key

K	Keyword — key types its BASIC word
L	Letter — lower-case typing
C	Caps lock (CAPS SHIFT + 2)
E	Extended — CAPS+SYMBOL shift, then key
G	Graphics — CAPS SHIFT + 9
In E mode a key gives the green word above it; with shift, the red word below.	

EDITING & KEYS

CAPS SHIFT +

Cursor ←→	CAPS SHIFT + 5 / 6 / 7 / 8
DELETE	CAPS SHIFT + 0
EDIT	CAPS SHIFT + 1
CAPS LOCK	CAPS SHIFT + 2
GRAPHICS	CAPS SHIFT + 9
BREAK	CAPS SHIFT + SPACE
SYMBOL SHIFT	Red symbols on the keys

USER-DEFINED GRAPHICS

UDG

Graphics mode	Keys A-U = 21 user characters
USR "a"	Address of the UDG "a" pattern
POKE USR "a"+r,b	Set row r (0-7) of the char
BIN 01111110	Handy for the 8 row bytes

MEMORY & SYSTEM VARS

16384 (\$4000)	Display file (pixels), 6144 bytes
22528 (\$5800)	Attributes (colour), 768 bytes
23552+	System variables
23606/7	CHARS — character-set pointer
23672-4	FRAMES — 1/50 s frame counter
23730/1	RAMTOP
48K RAM runs \$4000-\$FFFF; the bottom 16K of the map is ROM.	

TIMEX 2068 EXTRAS

vs Spectrum

SOUND r,d	AY-3-8912 register / data
STICK(n)	Read a joystick port
ON ERR	Trap an error
FREE	Free memory
DELETE	Delete a line range
RESET	Reset machine state
72K bank-switched + cartridge dock. BASIC is a Spectrum superset, but only ~7% of Spectrum machine-code runs without the emulator cartridge.	

QUICK-FACTS

Spectrum 16/48K	Z80A 3.5 MHz · ULA · BEEP · 1982
Spectrum 128	+ AY-3-8912 · 128K · RS232 · PLAY
TS2068 (1983)	Z80A 3.5 · 48K · AY sound · cartridges
Display	256×192 · 32×24 colour cells · 8 colours